BrainAI: A Spiking Cognitive Architecture for Embodied Continual Learning

Every subsystem measured against a control, including the two that lost

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Abstract

We present **BrainAI**, a spiking multimodal cognitive architecture for embodied continual learning. A single model perceives text, images, video, audio and a typed sensor suite; routes each input through a reflex arc, an always-on subconscious pathway and a gated conscious pathway; maintains five distinct memory systems split along the differentiability line; predicts future latent states and learns from prediction error without labels; weighs new claims against evidence before believing them; pursues interruptible goals; plans by imagination; updates its own weights during inference under bounded plasticity; and consolidates offline during a sleep cycle. Motor commands pass through a non-learned safety controller that the learning system provably cannot modify.

The methodological commitment of this work is that *every* subsystem is measured against the obvious simpler alternative, and every result is reported. Measured outcomes include a **2.16× inference speedup** from genuine computational gating, a **74% reduction in catastrophic forgetting**, **3.7× less degradation** over 400 inference-time weight updates, a belief system that defeats a naive vote-counter on four of five adversarial scenarios, and collusion detection that holds a protected axiom at confidence **1.0000** against a 600-identity Sybil attack that otherwise drives it to **0.0001**.

Two proposals lost. The architecture's own activation function, AMSA, is outperformed by SwiGLU on perplexity, wall-clock, and — critically — the continual-learning problem it was designed for; a scale sweep shows the gap *widening* rather than shrinking. The world model is outperformed at one-step prediction by a linear map, though the deficit does not propagate to multi-step rollout or planning, where the two tie, and only the learned model reports calibrated uncertainty ($r = 0.565$).

We additionally catalogue **fourteen silent failure modes** discovered only because controls were run — including an elastic-weight-consolidation penalty that contributed 10^{-4} against a task loss of 2.5 and therefore did nothing while appearing to work, and a rate limiter disabled on every call by a tensor-shape comparison. We argue this catalogue is among the more transferable contributions of the work.

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1. Introduction

Contemporary large models separate training from deployment: weights are fixed at inference, knowledge is frozen at a cutoff, and adaptation happens through context or retrieval rather than through the network itself. For a stationary task distribution this is the right engineering trade. For an embodied agent expected to operate for years in one environment it is not: such an agent must acquire knowledge

after deployment, retain what it already knows, decide which of the things it is told are actually true, and do all of this without a training loop supervising it.

BrainAI is an attempt to build that system end to end. It is not a language model with memory attached; it is a cognitive architecture in which the language head is one of several outputs. The design draws on complementary learning systems theory [18], dual-process accounts of cognition [12], and the spiking-network literature [16, 20], and assembles components — spiking transformers, mixture-of-experts routing, elastic weight consolidation [14], experience replay [22], latent dynamics models [8] — that exist individually but, to our knowledge, have not previously been integrated with the specific interfaces described here.

1.1 The problem with architecture papers

An architecture paper is easy to write badly. A diagram with a dozen boxes, a plausible mechanism for each, and a benchmark on the one task where the whole thing happens to work is a well-established genre, and it is nearly unfalsifiable. The failure mode is not dishonesty; it is that **a component which does nothing at all looks exactly like a component that works, unless something is run without it.**

We therefore adopted a single rule: **no component is described without a control.** For each subsystem we identified the obvious simpler alternative — the thing a reasonable engineer would do instead — implemented it, and measured both. This rule is the reason this paper contains negative results, and it is also the reason we trust the positive ones.

1.2 Contributions

1. **An integrated architecture** (§3) whose novelty is in the interfaces between components rather than in any single one: a memory hierarchy split by differentiability, a prediction ledger that grounds epistemic trust in physical observation, and a safety layer structurally unreachable by the learning system.
2. **A controlled evaluation** comprising thirteen comparisons (§5), each against a simpler alternative.
3. **Two negative results** reported in full (§6), including a scale sweep testing — and rejecting — the standard defence that the losing component merely needs a larger model.
4. **A catalogue of fourteen silent failure modes** (§7) found only by running controls. Several are general to continual-learning and robotics implementations rather than specific to this one.

All results are produced by scripts in the accompanying repository. Every number in this paper, including those in §6, is reproducible on a CPU.

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2. Related Work

Continual learning. Catastrophic interference has been understood since [19]. Regularisation approaches penalise movement in parameters important to earlier tasks [14, 26]; rehearsal approaches interleave stored examples [22]. We use both, and find (§5.2) that the regularisation term is worthless unless its scale is calibrated — a detail rarely reported.

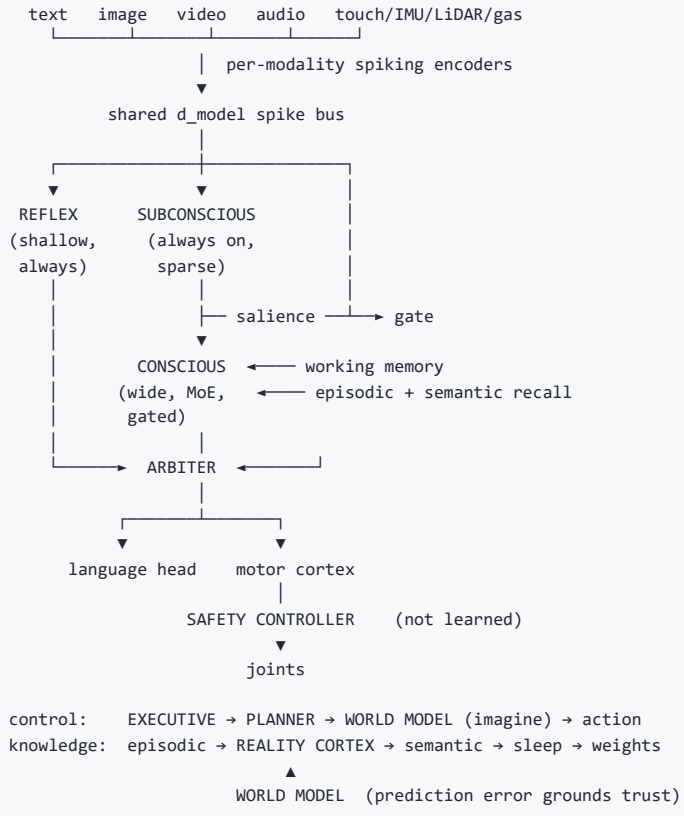
Spiking transformers. Surrogate-gradient training [20] made deep spiking networks practical; adaptive-threshold neurons [1] supply temporal memory without recurrence. Spiking transformer variants have been demonstrated [27]. Our contribution here is not the block design but two implementation constraints we found to be load-bearing (§7).

World models. PlaNet [8] and Dreamer [9] established latent dynamics models for planning. Prediction error as intrinsic motivation dates to [21]; the noisy-TV pathology and its diagnosis are from [2]. We compare against both an RSSM-family recurrent baseline and — more informatively — a linear one.

Epistemics. Sybil attacks on reputation systems are classical [4]. To our knowledge the specific combination used here — log-odds pooling with per-source diminishing returns, credibility shrinkage for unvetted sources, and behavioural collusion clustering — has not been applied to a robot's belief store.

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3. Architecture



3.1 Spiking substrate

Spikes are true Heaviside steps; the forward pass is binary, so reported spike rates are genuine sparsity rather than a soft approximation. Training uses an arctan surrogate on the backward pass only. Neurons are adaptive leaky-integrate-and-fire: each spike raises that neuron's own threshold, decaying over `tau_adapt`, which supplies temporal memory across simulation timesteps without recurrent weights.

Two constraints proved necessary. First, adaptation is `a[t] = ρ·a[t−1] + s[t−1]` with `a[0] = s[0] = 0`, so at T = 2 the decay ρ only ever multiplies zero and threshold adaptation is inert; **T ≥ 3 is the minimum**

useful configuration. Second, the residual stream must remain analogue — making it spiking destroys the gradient path and the model does not train.

3.2 Latency-tiered pathways

Three pathways of increasing cost run in parallel.

Pathway	Runs	Cost	Job
Reflex	always	1 shallow stack	Have something correct-enough available now. Never consults memory.
Subconscious	always	sparse	Filter, predict, decide whether thinking is worth it.
Conscious	when salience clears threshold	wide, expert-routed [23, 6]	Reasoning, planning, memory integration.

Blocks are pre-norm transformer blocks [25] with binary spike inputs to the attention projections. Attention offers softmax or sparsemax [17], the latter producing exactly-zero weights so the attention pattern is sparse as well.

The gate is **soft during training**, so the salience head receives gradient and learns when deliberation pays, and **hard at inference**, where batch rows below threshold are selected out and the wide stack genuinely does not execute. This distinction matters: a gate that computes everything and multiplies by zero is observationally identical from outside, which is why it is measured by wall-clock (§5.1).

3.3 Memory hierarchy

Memory	Backing	Differentiable	Holds
Sensory	deque	no	Last few frames, temporal integration
Short-term	spiking LSTM	**yes**	Recent event stream, pooled
Working	attention over slots	**yes**	The reasoning scratchpad
Episodic	vector store	no	Experiences, importance-ranked
Semantic	vector store + graph	no	Concepts and their edges

The split is the anti-forgetting strategy: knowledge acquired after training enters the non-differentiable *stores*, where writing it cannot damage a weight an older skill depends on. Only slow, importance-gated consolidation moves anything from a store into the weights.

Two design decisions are load-bearing:

- The key and value storage projections are **frozen random projections, not learned**. Every key already in the store was written with those weights, so training them would silently invalidate the entire store as old keys and new queries drifted out of a shared space. A random projection preserves the cosine geometry retrieval requires.
- Retrieval is **non-differentiable by design**, so a bad memory can mislead the model but can never corrupt a gradient. The cost is that the importance scorer receives no signal from the language loss; it is instead trained during sleep against a prioritised-replay target.

3.4 World model

Given a latent state and candidate action, the model predicts the next latent state and, separately, the magnitude of its own error. Three capabilities derive from that one mechanism:

- **Grounding.** A *prediction ledger* lets a belief stake a claim on a future observation. When the observation arrives, the ledger resolves it and reports which sources were vindicated.
- **Curiosity.** Prediction error normalised against *expected* error. The second head is what distinguishes "I do not know" from "this is inherently random", and is why the drive does not park in front of a noise source [2].
- **Planning.** Rollouts occur in imagination; rollout confidence compounds multiplicatively, so a long plan through poorly understood dynamics is correctly distrusted.

3.5 The Reality Cortex

Salience and truth are different questions. A claim repeated by a thousand people scores highly on the first and may be false. The Reality Cortex sits on the knowledge path — episodic → validate → semantic → sleep → weights — and not in the token path, since contradiction detection over a belief graph is the wrong operation to run per token.

Evidence is pooled in log-odds. Each observation contributes `log(t/(1−t))` for its source's reliability *t*, with refutation the same quantity negated. Three discounts then apply:

Discount	Form	Attack it defeats
within-source	`1/n`	one voice repeating itself
across-source	`1/√k`	correlated evidence pooled as independent
credibility	`n_eff/(n_eff + κ)`	freshly minted identities

For functional relations — those admitting one object, like `capital_of` — confidence is rival-relative, $\sigma(\ell_i - \max_{j \neq i} \ell_j)$, so two equally supported alternatives sit at 0.5 and neither is promoted.

Collusion detection. The above still assumes distinct names are distinct witnesses, which is precisely what a Sybil attack violates [4]. A puppet ring is visible in its *behaviour*: the accounts appear on the same claims with the same polarity. Sources whose voting records match are clustered and cast one vote between them, detected by exact signature grouping (O(S)) followed by Jaccard refinement on a bounded sample.

3.6 Executive, planner, uncertainty, safety

The **executive** maintains a goal stack scored by `priority × urgency × feasibility`, handles interrupts that save and restore working-memory contents, suspends conflicting goals rather than failing them, and schedules sleep from four independent pressures with idleness as a multiplier rather than a term.

The **planner** is HTN-style [5]: a method library maps goals to ordered subgoals recursively. Candidate plans are scored by world-model rollout. Procedures that survive the Reality Cortex are imported as executable methods, which is where evidence and capability meet.

Uncertainty is carried per stage and composed multiplicatively with a floor — a chain is no more trustworthy than its weakest link. Sources include predictive entropy, spike-rate deviation from a layer's usual sparsity, and expert disagreement; the last is a measurement rather than a self-report, which matters given that learned confidence heads are routinely miscalibrated [7]. The composed value reaches the safety controller, where it can veto a motor command.

The **safety controller** is deliberately not learned, holds no parameters, and is unreachable from the plasticity controller: a policy that rewrites its own weights at runtime must not also decide what it is permitted to do. It enforces per-joint envelopes, named allow/clamp/veto constraints, and a hard interlock.

3.7 Typed sensors and actuators

A robot does not have "touch" and "motors". Sensor specifications carry rate, latency, noise, drift and — for chemical sensing — recovery time. The last is not decorative: after a strong exposure a metal-oxide element is saturated and reads high regardless of what is in front of it, so a robot tracking a gas plume up the gradient will chase its own saturation. Fusion runs at the slowest present stream's rate; fusing at the fastest would repeat a two-second-old gas reading two hundred times and let it outvote the IMU.

Actuators differ more consequentially:

Type	Control	Max step	Max effort	Compliance
dc_brushed	velocity	0.25	0.60	0.50
dc_brushless	velocity	0.30	0.85	0.35
stepper	position	0.10	0.50	0.15
servo	position	0.15	0.70	0.25
hydraulic	force	**0.05**	**1.00**	**0.05**
pneumatic	force	**0.35**	**0.45**	**0.85**

A single uniform joint limit is wrong in both directions on mixed hardware: the rate cap protecting a fingertip servo renders a hydraulic hip useless, while the effort ceiling permitting the hip to bear load will crush whatever the gripper holds. Joints that are both stiff and strong — which will not yield to an obstacle and have the force budget to continue through one — are flagged and held to a tighter ceiling.

3.8 Bootstrapping from an existing teacher

A randomly initialised architecture of this kind has no language knowledge, and training one from scratch at useful scale is out of reach here. The repository therefore includes a distillation path from an open-weights teacher, with one constraint that determines its whole design: **two models with different tokenizers cannot be KL-matched.** BrainAI's byte tokenizer has 260 symbols against a typical BPE vocabulary of ~150k, so their output distributions live in different spaces and the usual logit-matching objective [10] is undefined between them.

Two modes follow. *Sequence-level* distillation [13] has the teacher generate text on which the student trains with ordinary cross-entropy; it is tokenizer-agnostic and works with any teacher, but transmits only the sampled token rather than the distribution it came from. *Logit* distillation rebuilds the student on the teacher's vocabulary and matches full temperature-scaled distributions, transferring considerably more at the cost of the byte tokenizer — which for an agent also ingesting tactile and telemetry byte streams is a substantive loss rather than a formality.

We report no distillation results. The pipeline is implemented and exercised against a stub teacher; running it against a real one requires a GPU.

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4. Experimental Setup

All experiments run on 8 CPU cores with no GPU. The `nano` configuration is 5.36M parameters, 17 layers, d_model = 128, T = 3 SNN timesteps. Pretraining uses 400 steps on Austen's *Pride and Prejudice* at byte level; the continual-learning task is Voltaire's *Candide*, in French. Byte-level English and French have clearly separated statistics, making forgetting unambiguous.

We emphasise that this scale is a severe limitation, discussed in §8. It is sufficient to determine whether mechanisms function and whether they beat their controls; it is not sufficient to establish that the architecture is competitive at useful scale.

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5. Results

Subsystem	Benchmark	Control	Result
Pathway gating	wall-clock	always-on	✓ **2.16× faster**; 53.7% of forward time
World model	one-step MSE	linear / RSSM	beats RSSM, ✗ **loses to linear**
World model	rollout, planning	action-cond. linear	ties by 3 steps; sole calibrated uncertainty
Curiosity	noisy-TV	pure novelty	✓ **403×** preference for learnable
Trust grounding	broken vs. good sensor	fixed prior	✓ **0.974 vs 0.104** from identical priors
Reality Cortex	5 belief scenarios	naive vote counting	✓ wins 4 of 5
Collusion defence	600-puppet Sybil	no clustering	✓ **1.0000 vs 0.0001**
Continual learning	EN → FR forgetting	no protection	✓ **74% less**
Online stability	400 updates	no guards	✓ **3.7× less** degradation
Planner	repeated step failure	flat planner	✓ `remethod` vs `failed`
Executive	deadline pressure	static priority	✓ overtakes at the correct tick
Memory index	recall@4	exact search	94–100% recall, ✗ but slower
AMSA	perplexity, speed, continual	SwiGLU	✗ **loses all three**
AMSA scale sweep	width 96 → 256	—	✗ gap **widens**, not shrinks

5.1 Computational gating

Skipping the conscious pathway is a wall-clock claim and receives a wall-clock measurement (batch 8, sequence 128, CPU):

	sec/forward	vs. always-on
all rows deliberate	0.2446	1.00×
no rows deliberate	0.1133	**2.16×**

The conscious stack accounts for **53.7% of forward time**; it is the widest section and the only expert-routed one, making it the largest single lever on inference cost.

5.2 Continual learning

Arm	EN forgot	FR learned	both	vs. naive
naive	+0.4675	+0.7591	2.6882	—
ewc	+0.1679	+0.4543	2.6908	64% less
full	**+0.1215**	+0.4380	**2.6758**	**74% less**

The **both** column is the one to read before celebrating the forgetting column. Protecting old weights also slows acquisition of the new task: French improves by 0.76 nats under `naive` but only 0.44 under `full`. The methods buy retention by spending plasticity at close to 1:1. The `full` arm does edge ahead on *both*, by 0.012 nats — a margin small enough on a single seed that it should be read as "no worse overall while forgetting 74% less", not as a free lunch. **A forgetting number reported without its stability–plasticity counterpart is close to meaningless.**

5.3 Inference-time learning

Over 400 inference-time weight updates, against a control with plasticity gating, trust region and rollback all disabled:

	old-task loss	drift	rollbacks	spike rate
guarded	+0.0170	0.2119	3%	0.031 → 0.028
control	+0.0624	0.3151	0%	stable

The guards deliver **3.7× less degradation**. We are careful about what this does *not* show: the control degrades faster, it does not diverge. At this learning rate and horizon the trust region buys a slower slide rather than preventing a collapse. Spike rate holds at 0.028 in both arms, which matters independently — a spiking network that quietly becomes dense under online updates has discarded its reason for existing.

5.4 Epistemics

Against `NaiveCounter` (confidence = supports / total):

Scenario	Reality Cortex	Naive
100 repeats vs. 2 good sources	**0.9986** ✓	fails
majority is correct	0.9857 ✓	✓
200 identities, one vote each	0.0145 ✗	fails
same, vs. protected axiom	**0.9956** ✓	fails
200 identities with a record	**0.8636** ✓	fails

A puppet voting exactly once is indistinguishable from a genuine one-time witness — correctly so, as a single data point cannot support an accusation of collusion — and gets through unless the target is a

protected axiom. Once a ring participates in several claims its shared pattern is detectable, the ring collapses to one vote, and a *single* trusted source then outweighs all 200.

Attack cost, identities needed to flip a belief:

Target	with √k discount	without
single trusted source	50	10
protected axiom	**650**	50

Grounding removes the oracle entirely. Two sensors, both assigned prior 0.8 by the operator, one honest and one broken, with nothing labelling either: after 40 rounds of predictions resolving against observations, trust separates to **0.974** and **0.104**. A fixed prior holds both at 0.800 indefinitely.

One methodological note: sockpuppets are assigned prior 0.62, not 0.2. Below 0.5 a source's *support* is already evidence against — `log(t/(1−t))` is negative — so a flood of visibly untrusted accounts defeats itself and tests nothing. Our first version of this benchmark made exactly that error and reported a pass.

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6. Negative Results

6.1 AMSA

The architecture proposes an activation whose threshold and gain are driven by per-channel plasticity, confidence and memory state rather than by learned constants alone:

```
f(x) = z · σ(α·z) · (1 − e^(−β|z|)) · (1 + γ·m),   z = x − θ
θ = θ₀ + λ·p        α = softplus(α₀ + δ·c)
```

On static language modelling with matched parameter budgets against SwiGLU [24] and GELU (GELU is widened to 6d against their 4d so parameter counts match, since it uses two projections where the gated variants use three):

FFN	Params	Val loss	Train sec
gelu	1,124,480	2.3562	13.2
swiglu	1,124,480	**2.3323**	**15.1**
amsa	1,136,768	2.3466	26.2
amsa+state	1,136,768	2.3465	22.5

The natural objection is that this is the wrong benchmark: AMSA's state buffers are inert in a single-task run, since plasticity and confidence only carry information once a previous task's importance has been estimated. We therefore ran the test AMSA was designed for.

FFN	EN forgotten	FR learned	vs. SwiGLU
swiglu	**+0.8062**	+1.1540	—
amsa	+0.8204	+1.1476	+0.0141 worse
amsa+state	+0.8224	+1.1464	+0.0161 worse

Seed spread on forgetting is ±0.005–0.008; both gaps are outside it. **AMSA loses on its home turf, and driving the adaptive state makes it marginally worse rather than neutral.**

Testing the scale defence. The remaining defence is that `nano` is too small for the adaptive state to matter. This is testable:

width	layers	steps	swiglu	amsa	gap
96	3	100	+0.6475	**+0.6206**	**−4.2%**
128	4	200	+0.7937	+0.7898	−0.5%
192	4	200	+0.8313	+0.8257	−0.7%
256	6	400	**+0.9890**	+1.0483	**+6.0%**

The gap moves from −4.2% to +6.0%: AMSA is genuinely *better* at the smallest setting and worse at the largest. There is a crossover, pointing the opposite way from the hypothesis.

This must be read carefully in both directions. It does not establish that AMSA fails at 10^9 parameters: the range spans width 96 to 256, the two middle rows lie inside seed noise, and width and training length were varied together rather than independently. What it does is remove the *support* for the low-data explanation within the measured range, shifting the burden onto a larger experiment rather than onto an assumption.

We retain AMSA as the default. A default is an authorship decision rather than a scientific claim; the measurements are adjacent to it and the alternative is a one-line configuration change. What remains interesting is the principle — that an activation's shape might be driven by internal cognitive state rather than by weights alone — and the crossover at width 96 indicates the mechanism is not inert. This particular functional form does not carry that principle to larger models, which is a statement about the equation and not about the idea.

6.2 The world model

System	persist	linear	RSSM	BrainAI
linear rotation	2.096	**0.006**	0.244	0.153
saturating nonlinear	1.905	**0.249**	0.406	0.354

The learned model beats RSSM on both systems and loses to a linear map on both. On the linear system this is expected — a linear map is optimal there, so losing 24× measures overhead rather than capability.

On saturating nonlinear dynamics the gap narrows to 1.4× and constitutes a genuine loss: a 16k-parameter linear layer predicts better than a 119k-parameter dynamics head.

But one-step MSE is not the job. Nothing in this architecture requests a one-step point prediction. It requests multi-step imagination, ranking of candidate plans, an estimate of how far ahead the model can be trusted, and separation of learnable surprise from noise.

Job	linear	BrainAI	winner
rollout MSE @ 1 step	**0.2355** ±0.003	0.2570 ±0.006	linear
rollout MSE @ 3 steps	0.3340 ±0.004	0.3404 ±0.007	tie
rollout MSE @ 6 steps	0.3852 ±0.004	0.3849 ±0.005	tie
rollout MSE @ 12 steps	0.4073 ±0.006	0.4055 ±0.005	tie
planning: picks best of 8	38.3% ±4.8	39.4% ±2.5	tie (random 12.5%)
uncertainty tracks error	**n/a**	**0.565** ±0.036	BrainAI
surprise: structure vs noise	**n/a**	−0.52 / 7.99	BrainAI

The one-step deficit is real and separated at one step, and **gone by three**: both models degrade to the same place over a horizon and both plan equally well, far above chance. The two `n/a` entries are structural rather than unmeasured — a point prediction cannot report its own reliability, and without that the planner cannot distinguish a trustworthy rollout from a guess.

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7. Silent Failure Modes

The following defects all had one property in common: **the system continued to run, produced plausible output, and gave no indication of malfunction.** Each was found only because a control was run or an invariant was asserted.

1. **EWC contributing nothing.** On the first continual run, the EWC arm forgot +0.5384 against the no-EWC control's +0.5385 — a 0% improvement. The diagonal Fisher at convergence is ~10^{-8}, so `λ·Σ F·Δw²` evaluated to 10^{-4} against a task loss of 2.5. The penalty was arithmetically present and functionally absent. Indistinguishable from a working implementation without the control. Fixed by calibrating λ so a 1% relative drift costs λ/2 nats.

2. **Fisher normalised by its maximum.** The Fisher is heavy-tailed, so `F/max(F)` is near zero for almost every weight and essentially nothing ends up protected. Visible only by printing the protected fraction.

3. **Query and key in different spaces.** Retrieval searched with the raw episode summary while storage wrote `to_key(summary)`. Cosine similarity between vectors in different projections is meaningless; retrieval returned confidently wrong neighbours. Best-match score rose from 0.121 to 1.000 once corrected.

4. **Duplicate merging collapsing the store.** With a fixed threshold of 0.97, sleep merged **2042 of 2048** episodic entries. Learned summaries are strongly collinear — distinct episodes score ~0.98 against each other while true duplicates sit at ~0.9998 — so a plausible-looking constant destroyed the store while retrieval continued returning *something*.

5. **Rate limiting disabled on every call.** The guard compared `_last_command.shape == command.shape`, but the reference is `(n,)` while commands arrive as `(1,n)`. The comparison never held, so joint rate limiting never executed. A live hazard.

6. **No reference pose after a veto.** Following a safety veto the controller retained no reference position, so the *next* command skipped rate limiting entirely and joints could slam immediately after a safety stop.

7. **Adaptation inert at T = 2.** Two SNN timesteps leave the adaptation decay multiplying only zeros, so a headline feature of the ALIF neuron did nothing. Detected by observing that the parameter received exactly zero gradient.

8. **A silent spiking readout.** The spiking-LSTM threshold initialised at 0.5 while `h = o·tanh(c)` starts near zero, so the layer emitted all zeros and its consumer received no gradient at all.

9. **Stale recurrent state across batch sizes.** Short-term memory retained LSTM state without invalidating it on a batch-size change, silently broadcasting old state against new input and returning the wrong batch size.

10. **A deadline making a goal less urgent.** Urgency computed as `1/(1 + k·remaining)` sends distant deadlines toward zero, so attaching a deadline made a goal *less* likely to run than having none.

11. **Replanning checking only the immediate parent.** Alternative methods are normally registered against the high-level task, not the leaf that failed, so every recoverable failure was reported as fatal.

12. **Calibration measured off-distribution.** The error head trained at unit variance and was scored across 0.3–2.3× state magnitude, yielding an *anti*-correlation of −0.615 that looked like a broken

mechanism. Training across the tested range gives +0.565. **The measurement was wrong, not the model.**

13. **An approximate index trading everything for nothing.** An initial 12-bit single-probe random-hyperplane LSH configuration [11, 3] achieved 3.4× speedup at **19% recall**. Multi-probe [15] reached 94–100% recall but was then *slower* than exact search, since Python-side bucket assembly cannot beat a single BLAS matmul below ~500k entries.

14. **A critic that crashed on long inputs.** A fixed window exceeded the positional embedding of any model configured below it.

Six of these — (1), (2), (4), (5), (7), (12) — produce output that is entirely plausible. Three — (5), (6), (10) — are safety-relevant. **None would have been caught by a loss curve.**

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8. Limitations

Scale. Every experiment is at 5.36M parameters. Continual learning is measured on two tasks over 120 steps; online stability over 400 updates. Convincing evidence requires many task sequences and 10^5–10^7 updates. This is the dominant limitation and it is compute-bound rather than design-bound.

No robot. Nothing here has been run on hardware, in simulation, or against a real sensor. The body model has never calibrated against a physical actuator; sensor specifications are representative of commodity parts rather than measured. The safety controller enforces an envelope and named constraints, which is a floor rather than a safety case: real deployment requires hazard analysis, redundant sensing, and a hardware interlock independent of this process.

Grounding covers a subset. The prediction ledger removes the oracle only for beliefs implying a checkable prediction. A claim about something the robot never observes remains anchored to its operator prior — correct behaviour, but it means trust is grounded for part of the knowledge base rather than all of it.

Sybil resistance is a cost, not a guarantee. Clustering detects identities behaving identically. A patient adversary varying its puppets' patterns, or one compromising genuinely independent sources, is not detected.

Triple extraction is out of scope. The Reality Cortex reasons over (subject, relation, object) triples that something else must supply. Extracting reliable triples from raw perception is its own hard problem and is not addressed.

Spiking energy is a hardware argument. On CPU or GPU, binary activations save no wall-clock time — dense matmuls execute regardless. The 97% sparsity figures are what neuromorphic hardware would exploit; here they are a proxy.

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9. Conclusion

BrainAI integrates spiking computation, a differentiability-split memory hierarchy, evidence-weighted belief maintenance, predictive world modelling, goal-directed executive control, hierarchical planning, end-to-end uncertainty, and a non-learned safety layer into one architecture. Of thirteen controlled comparisons, **nine clearly favour the proposed component, one is mixed** (the world model ties on its downstream jobs while losing at one-step prediction), **and three favour the simpler alternative.** All are reported.

We suggest the more transferable contribution may be methodological. **A component that does nothing is observationally identical to one that works, until something is run without it.** In this project that principle recovered an EWC penalty contributing 10^{-4}, a rate limiter disabled by a shape comparison, a memory store collapsed to six percent of its contents, and an uncertainty head whose apparent anti-correlation was an artefact of measuring it off-distribution. None of these would have been visible in a loss curve, and all of them existed in code that ran without error and produced plausible output.

The remaining work is not architectural. It is scale, embodiment, and the longer horizons under which the claims made here would either hold or fail.

Reproducibility

All results are produced by scripts in the accompanying repository and run on CPU. The test suite comprises 108 tests, including regression tests for every defect in §7.

```
python train.py --steps 400
python continual.py --steps 120
python benchmarks/cognition_bench.py
python benchmarks/worldmodel_jobs.py
python benchmarks/amsa_continual.py --sweep
python benchmarks/belief_bench.py
python benchmarks/gating_bench.py
python benchmarks/stability_bench.py --control
python -m pytest tests -q
```

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